

Dirac delta function

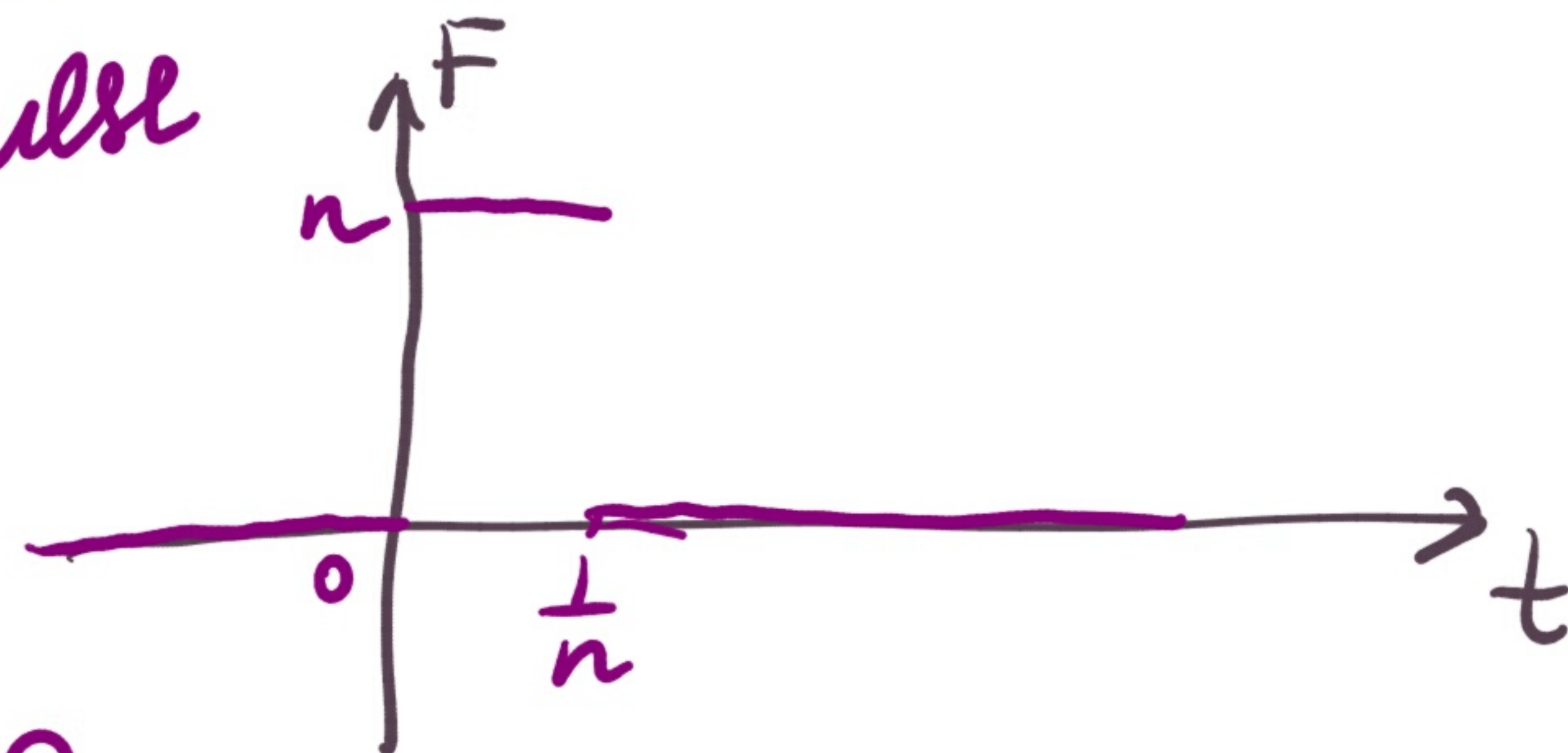
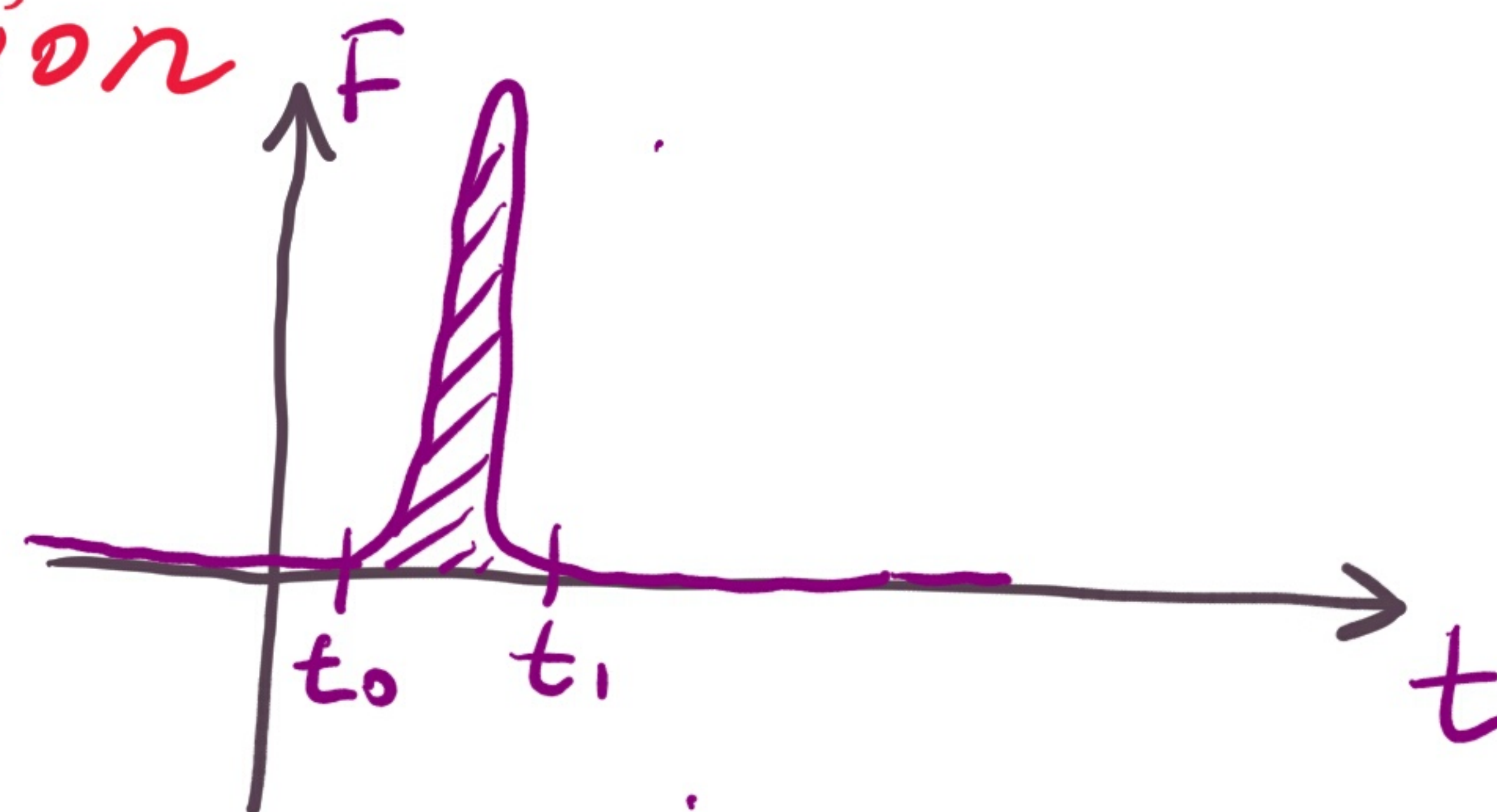
$$ay'' + by' + cy = F(t) \leftarrow \text{ext. force}$$

$$mV(t_1) - mV(t_0) = \int_{t_0}^{t_1} m \frac{dV}{dt} dt = \int_{t_0}^{t_1} F(t) dt$$

$\uparrow$
 change in momentum

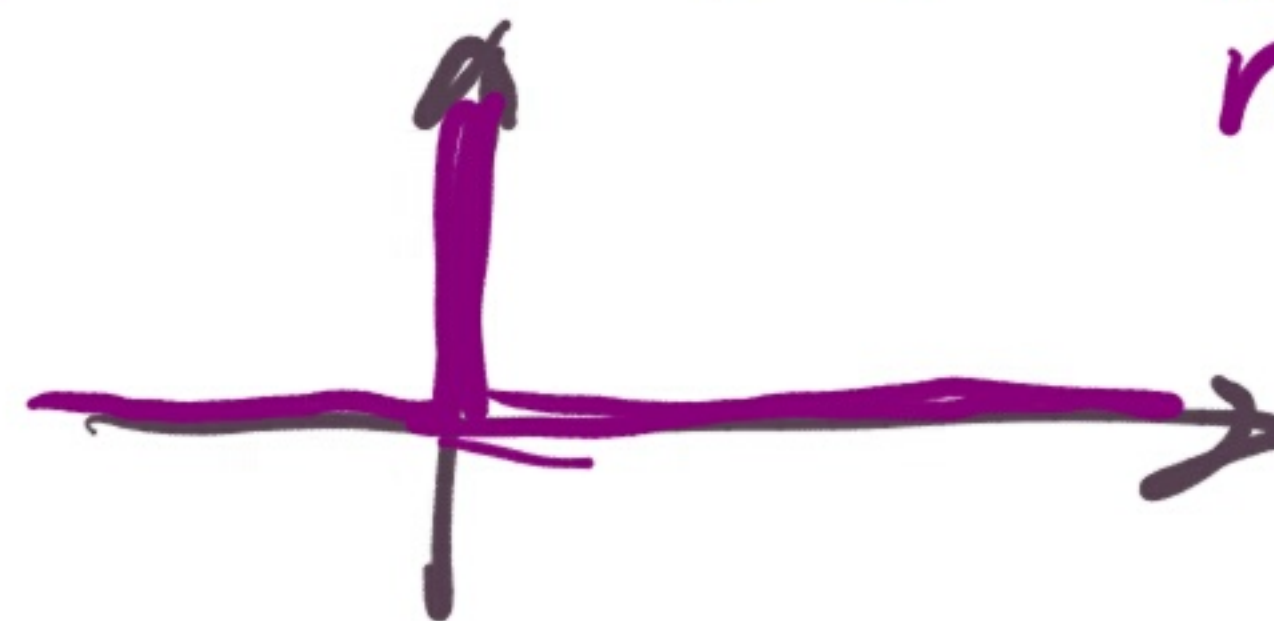
$\underbrace{\int_{t_0}^{t_1} F(t) dt}_{\text{impulse}}$

$$F_n(t) = \begin{cases} n, & 0 < t < \frac{1}{n} \\ 0, & \text{otherwise} \end{cases}$$



Def: Dirac delta function δ as $\lim_{n \rightarrow \infty} F_n(t)$

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$$



$$\int_{-\infty}^{\infty} \delta(t) dt = \int_{-\infty}^{\infty} \lim_{n \rightarrow \infty} F_n(t) dt = \int_{-\infty}^{\infty} \lim_{n \rightarrow \infty} \begin{cases} n, & 0 < t < \frac{1}{n} \\ 0, & \text{otherwise.} \end{cases} dt$$

$$= \lim_{n \rightarrow \infty} \int_0^{1/n} n dt = \lim_{n \rightarrow \infty} [nt]_0^{1/n} = 1$$

$$\bullet \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\bullet \int_{-\infty}^{\infty} f(t) \delta(t) dt = \lim_{n \rightarrow \infty} \int_0^{1/n} n f(t) dt =$$

$$= \lim_{n \rightarrow \infty} n \cdot f(c_n) \cdot \frac{1}{n} = \lim_{n \rightarrow \infty} f(c_n) = f(0)$$

$c_n \in (0, \frac{1}{n})$

$$\bullet \int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0)$$

mean value thm:

$$\int_a^b g(t) dt = g(c)(b-a)$$

$c \in (a, b)$

$$\bullet \int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0) \checkmark = f(t)$$

$$\mathcal{L}\{\delta\}(s) = \int_{-\infty}^{\infty} \underbrace{e^{-st}}_{=f(t)} \delta(t) dt = e^{-s \cdot 0} = 1$$

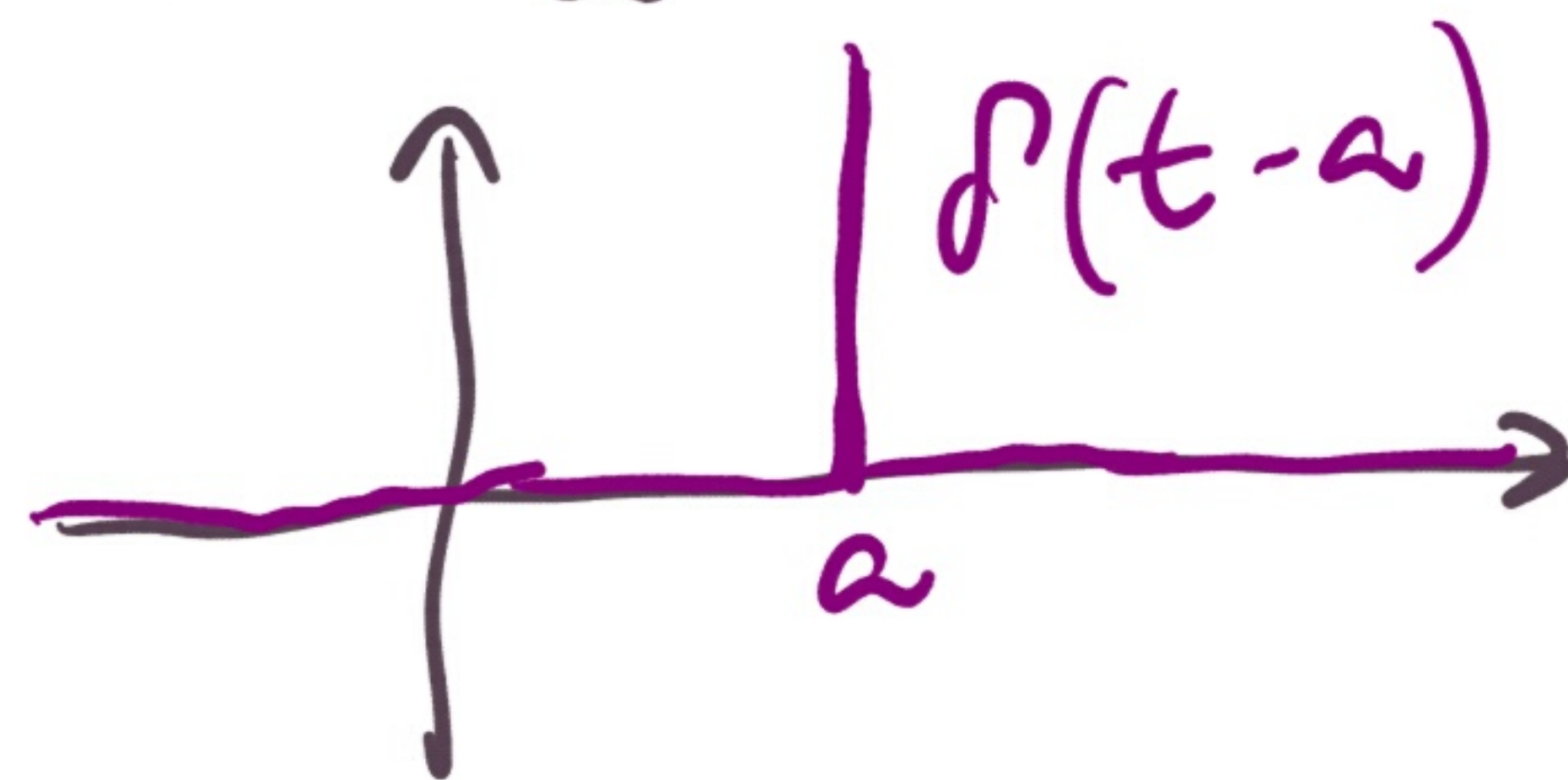
$$\bullet \mathcal{L}\{\delta\}(s) = 1$$

$$\bullet \int_{-\infty}^{\infty} \delta(t-a) dt = \left(\begin{array}{l} u = t - a \\ du = dt \end{array} \quad u \Big|_{-\infty}^{\infty} \right) = \int_{-\infty}^{\infty} \delta(u) du = 1$$

$$\bullet \int_{-\infty}^{\infty} f(t) \delta(t-a) dt = \int_{-\infty}^{\infty} f(u+a) \delta(u) du =$$

$$= f(0+a) = f(a)$$

$$\bullet \int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)$$



$$\mathcal{L}\{f(t)\delta(t-a)\}(s) = \int_{-\infty}^{\infty} \underbrace{e^{-st} f(t)}_{\delta(t-a)} dt =$$

$$= e^{-sa} f(a)$$

- $\int_{-\infty}^{\infty} \delta(t) dt = 1$

- $\int_{-\infty}^{\infty} f(t) \delta(t) dt = f(0)$

- $\int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)$

- $\mathcal{L}\{\delta\}(s) = 1$

- $\mathcal{L}\{f(t)\delta(t-a)\}(s) =$
 $= e^{-sa} f(a)$

Ex: $y'' + y = \delta(t)$, $y(0) = 0, y'(0) = 0$

$\delta(t) = \lim_{n \rightarrow \infty} F_n(t)$, where $F_n(t) = \begin{cases} n, & 0 < t < \frac{1}{n} \\ 0, & \text{oth.} \end{cases}$

$y_n'' + y_n = F_n(t)$, $y_n(0) = 0, y_n'(0) = 0$

$s^2 Y_n(s) + Y_n(s) = \mathcal{L}\{n - n u(t - \frac{1}{n})\}(s)$

$Y_n(s)(s^2 + 1) = \frac{n}{s} - n \cdot \frac{e^{-s \cdot \frac{1}{n}}}{s} = \frac{n}{s} (1 - e^{-\frac{s}{n}})$

$Y_n(s) = \frac{n}{s(s^2 + 1)} - n \frac{e^{-\frac{s}{n}}}{s(s^2 + 1)} \quad \left(\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1} \right)$

$y_n(t) = n - n \cos t - u(t - \frac{1}{n}) [n - n \cos(t - \frac{1}{n})]$

$$y_n(t) = n - n \cos t - [n - n \cos(t - \frac{1}{n})] u(t - \frac{1}{n})$$

Fix $t > 0$, assume that $\frac{1}{n} < t$, so $u(t - \frac{1}{n}) = 1$

$$\Rightarrow y_n(t) = \cancel{n} - n \cos t - \cancel{n} + n \cos(t - \frac{1}{n}) =$$

$$= n \cos(t - \frac{1}{n}) - n \cos t$$

$$\lim_{n \rightarrow \infty} y_n(t) = \lim_{n \rightarrow \infty} (n \cos(t - \frac{1}{n}) - n \cos t) =$$

$$= \lim_{n \rightarrow \infty} \frac{\cos(t - \frac{1}{n}) - \cos t}{\frac{1}{n}} = \left(h = \frac{1}{n} \right) =$$

$$= \lim_{h \rightarrow 0} \left(- \frac{\cos(t+h) - \cos t}{h} \right) = - \lim_{h \rightarrow 0} \frac{\cos(t+h) - \cos t}{h} =$$

$$= - \frac{d}{dt} (\cos t) = \sin t$$

limit of sol. y_n of $y_n'' + y_n = F_n(t)$, $y_n(0) = y_n'(0) = 0$
is $\boxed{\sin t}$

We will assume that $\boxed{\sin t}$ is sol of $y'' + y = \delta(t)$, $y(0) = y'(0) = 0$

Solve $y'' + y = \delta(t)$, $y(0) = y'(0) = 0$

$$s^2 Y(s) + Y(s) = \mathcal{L}\{\delta\}(s) = 1$$

$$Y(s) = \frac{1}{1+s^2}$$

$$\Rightarrow \boxed{y(t) = \sin t}$$

Ex: $y'' + 9y = 3\delta(t - \pi)$, $y(0) = 1$, $y'(0) = 0$

$$s^2 Y(s) + 9Y(s) = 3 \mathcal{L}\{\delta(t - \pi)\}(s) = 3e^{-\pi s}$$

$$Y(s) = 3 \frac{e^{-\pi s}}{s^2 + 9} \Rightarrow y(t) = 3 \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2 + 9}\right\}(t) =$$

$$= u(t - \pi) \mathcal{L}^{-1}\left\{\frac{3}{s^2 + 3^2}\right\}(t - \pi) =$$

$$= u(t - \pi) \sin 3(t - \pi) = u(t - \pi) \sin(3t - \pi) =$$

$$= -u(t - \pi) \sin 3t$$

Ex: $ay'' + by' + cy = \delta(t)$, $y(0) = y'(0) = 0$ (1)

$$as^2 Y(s) + bs Y(s) + cY(s) = 1$$

$$Y(s) = \frac{1}{as^2 + bs + c} = H(s)$$

$$y(t) = \mathcal{L}^{-1}\{H\}(t) = h(t) \leftarrow \begin{matrix} \text{impulse} \\ \text{response} \\ \text{function} \end{matrix}$$

To solve (1) find $h(t) = y(t)$.

Ex: $y'' - y = \delta(t)$, $y(0) = y'(0) = 0$

$$\begin{aligned} \Rightarrow h(t) &= \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 1}\right\}(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)(s+1)}\right\}(t) = \\ &= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1} - \frac{1}{s+1}\right\}(t) = \frac{1}{2}(e^t - e^{-t}) \Rightarrow y(t) = \frac{1}{2}(e^t - e^{-t}) \end{aligned}$$